∑ StackEditor Formula Reference Catalog

Comprehensive All-Level Mathematical, Physical, Chemical, Biological & Statistical Equation Library

V1.1.0 OFFICIAL

1. Basic Mathematical Structures, Radicals & Notations

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Fraction Standard fraction	$\frac{a}{b}$	`\frac{a}{b}`
Differential Fraction Derivative ratio	$\frac{dy}{dx}$	`\frac{dy}{dx}`
Square Root Square root of x	$\sqrt{x}$	`\sqrt{x}`
N-th Root N-th radical root	$\sqrt[n]{x}$	`\sqrt[n]{x}`
Superscript / Power Exponent power	x^{n}	`x^{n}`
Subscript Index subscript	x_{i}	`x_{i}`
Sub & Super Both index and power	x_{i}^{n}	`x_{i}^{n}`
Absolute Value Absolute value / modulus	$\left\| x \right\|$	`\left| x \right|`
Vector Norm Magnitude / norm	$\left\\| \vec{v} \right\\|$	`\left\\| \vec{v} \right\\|`
Floor Function Floor bracket	$\lfloor x \rfloor$	`\lfloor x \rfloor`
Ceiling Function Ceiling bracket	$\lceil x \rceil$	`\lceil x \rceil`
Dynamic Parentheses Auto-sizing round parentheses	$\left(\frac{a}{b} \right)$	`\left( \frac{a}{b} \right)`
Dynamic Brackets Auto-sizing square brackets	$\left[\frac{a}{b} \right]$	`\left[ \frac{a}{b} \right]`
Dynamic Braces Auto-sizing curly braces	$\left\{ x \in \mathrm{R} \mid x > 0 \right\}$	`\left\{ x \in \mathbb{R} \mid x > 0 \right\}`
Angle Brackets (Bra-Ket) Inner product / Bra-ket	$\langle u, v \rangle$	`\langle u, v \rangle`
Piecewise / Cases Piecewise conditional function	$f(x) = \begin{cases} x^2 & x \ge 0 \\ -x & x < 0 \end{cases}$	`f(x) = \begin{cases} x^2 & x \ge 0 \\ -x & x < 0 \end{cases}`
Overline / Conjugate Complex conjugate or mean	$\overline{z}$	`\overline{z}`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Vector Arrow Vector arrow over letter	$\vec{v}$	`\vec{v}`
Unit Vector (Hat) Unit vector / estimator	$\hat{u}$	`\hat{u}`
Time Derivative (Dot) First time derivative	$\dot{x}$	`\dot{x}`
Second Time Derivative (Ddot) Second time derivative	$\ddot{x}$	`\ddot{x}`

2. Algebra, Factoring, Series, Finance & Conic Geometry

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Simple Interest Simple interest calculation	$I = P \cdot r \cdot t$	`I = P \cdot r \cdot t`
Compound Interest Compound interest with periodic compounding	$A = P \left(1 + \frac{r}{n}\right)^{n t}$	`A = P \left(1 + \frac{r}{n}\right)^{n t}`
Difference of Squares Algebraic factoring identity	$a^2 - b^2 = (a - b)(a + b)$	`a^2 - b^2 = (a - b)(a + b)`
Perfect Square Trinomial Binomial expansion identity	$(a \pm b)^2 = a^2 \pm 2 a b + b^2$	`(a \pm b)^2 = a^2 \pm 2 a b + b^2`
Sum & Difference of Cubes Cubic factoring identity	$a^3 \pm b^3 = (a \pm b)(a^2 \mp a b + b^2)$	`a^3 \pm b^3 = (a \pm b)(a^2 \mp a b + b^2)`
Slope-Intercept Line Standard slope-intercept form	$y = m x + b$	`y = m x + b`
Slope Formula Rate of change between coordinates	$m = \frac{y_2 - y_1}{x_2 - x_1}$	`m = \frac{y_2 - y_1}{x_2 - x_1}`
Point-Slope Equation Line passing through (x1, y1)	$y - y_1 = m (x - x_1)$	`y - y_1 = m (x - x_1)`
Quadratic Formula Roots of $ax^2 + bx + c = 0$	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	`x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}`
Quadratic Discriminant Determines nature of quadratic roots	$\Delta = b^2 - 4 a c$	`\Delta = b^2 - 4 a c`
Parabola Vertex Form Parabola with vertex at (h, k)	$y = a (x - h)^2 + k$	`y = a (x - h)^2 + k`
2D Distance Formula Euclidean distance between points	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	`d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}`
Midpoint Formula Coordinates of line midpoint	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$	`M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Equation of Circle Circle center (h, k), radius r	$(x - h)^2 + (y - k)^2 = r^2$	`(x - h)^2 + (y - k)^2 = r^2`
Equation of Ellipse Standard conic section ellipse	$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$	`\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1`
Equation of Hyperbola Standard conic section hyperbola	$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$	`\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1`
Pythagorean Theorem Right triangle hypotenuse relation	$a^2 + b^2 = c^2$	`a^2 + b^2 = c^2`
Triangle Area Base times height area	$A = \frac{1}{2} b h$	`A = \frac{1}{2} b h`
Heron's Formula Triangle area from 3 sides	$A = \sqrt{s(s - a)(s - b)(s - c)}, \quad s = \frac{a + b + c}{2}$	`A = \sqrt{s(s - a)(s - b)(s - c)}, \quad s = \frac{a + b + c}{2}`
Trapezoid Area Area of a trapezoid	$A = \frac{a + b}{2} h$	`A = \frac{a + b}{2} h`
Area & Circumference of Circle Circular geometry formulas	$A = \pi r^2, \quad C = 2 \pi r$	`A = \pi r^2, \quad C = 2 \pi r`
Sphere Volume & Surface Area 3D sphere spatial formulas	$V = \frac{4}{3}\pi r^3, \quad A = 4 \pi r^2$	`V = \frac{4}{3}\pi r^3, \quad A = 4 \pi r^2`
Cylinder Volume Right circular cylinder volume	$V = \pi r^2 h$	`V = \pi r^2 h`
Cone Volume Right circular cone volume	$V = \frac{1}{3}\pi r^2 h$	`V = \frac{1}{3}\pi r^2 h`
Binomial Theorem Polynomial power expansion	$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$	`(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k`
Arithmetic Sequence & Sum N-th term and sum of arithmetic series	$a_n = a_1 + (n - 1)d, \quad S_n = \frac{n}{2}(a_1 + a_n)$	`a_n = a_1 + (n - 1)d, \quad S_n = \frac{n}{2}(a_1 + a_n)`
Geometric Sequence & Series Finite and infinite geometric series sum	$a_n = a_1 r^{n-1}, \quad S_n = a_1 \frac{1 - r^n}{1 - r}, \quad S_\infty = \frac{a_1}{1 - r}$	`a_n = a_1 r^{n-1}, \quad S_n = a_1 \frac{1 - r^n}{1 - r}, \quad S_\infty = \frac{a_1}{1 - r}`
Euler's Identity & Formula Fundamental mathematical identity	$e^{i \pi} + 1 = 0 \iff e^{i \theta} = \cos\theta + i \sin\theta$	`e^{i \pi} + 1 = 0 \iff e^{i \theta} = \cos\theta + i \sin\theta`
De Moivre's Formula Powers of complex numbers	$(\cos\theta + i \sin\theta)^n = \cos(n\theta) + i \sin(n\theta)$	`(\cos\theta + i \sin\theta)^n = \cos(n\theta) + i \sin(n\theta)`
Combinations & Permutations Combinatorics counting formulas	$\binom{n}{k} = \frac{n!}{k!(n - k)!}, \quad P(n,k) = \frac{n!}{(n - k)!}$	`\binom{n}{k} = \frac{n!}{k!(n - k)!}, \quad P(n,k) = \frac{n!}{(n - k)!}`

3. Calculus, Series Expansions, Differential Equations & Vector Calculus

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Derivative Definition (Limit) Formal difference quotient	$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$	`f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}`
L'Hôpital's Rule Indeterminate limit evaluation	$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$	`\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}`
Product Rule Derivative of product of functions	$\frac{d}{dx}[u \cdot v] = u' v + u v'$	`\frac{d}{dx}[u \cdot v] = u' v + u v'`
Quotient Rule Derivative of fraction of functions	$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u' v - u v'}{v^2}$	`\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u' v - u v'}{v^2}`
Chain Rule Composite function differentiation	$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$	`\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)`
Integration by Parts Integral of product of functions	$\int u \, dv = u v - \int v \, du$	`\int u \, dv = u v - \int v \, du`
Fundamental Theorem of Calculus Connection between integral and antiderivative	$\int_{a}^{b} f(x) \, dx = F(b) - F(a), \quad F'(x) = f(x)$	`\int_{a}^{b} f(x) \, dx = F(b) - F(a), \quad F'(x) = f(x)`
Arc Length of Curve Curve arc length integration	$L = \int_{a}^{b} \sqrt{1 + \left[f'(x)\right]^2} \, dx$	`L = \int_{a}^{b} \sqrt{1 + \left[f'(x)\right]^2} \, dx`
Volume of Revolution (Disc) Rotational solid volume	$V = \pi \int_{a}^{b} [f(x)]^2 \, dx$	`V = \pi \int_{a}^{b} [f(x)]^2 \, dx`
Taylor Series Expansion Infinite polynomial approximation	$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$	`f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n`
Maclaurin Exponential Series Exponential power series	$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	`e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots`
Fourier Transform Continuous Fourier frequency transform	$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} \, dx$	`\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} \, dx`
Laplace Transform Complex frequency domain transform	$\mathcal{L}\{f(t)\}(s) = \int_{0}^{\infty} f(t) e^{-s t} \, dt$	`\mathcal{L}\{f(t)\}(s) = \int_{0}^{\infty} f(t) e^{-s t} \, dt`
First-Order Linear ODE Integrating factor ODE solution	$\frac{dy}{dx} + P(x) y = Q(x) \implies y = e^{-\int P dx} \left(\int Q e^{\int P dx} dx + C \right)$	`\frac{dy}{dx} + P(x) y = Q(x) \implies y = e^{-\int P dx} \left( \int Q e^{\int P dx} dx + C \right)`
Simple Harmonic Oscillator ODE Second-order harmonic oscillation ODE	$\frac{d^2 y}{dt^2} + \omega^2 y = 0 \implies y(t) = A \cos(\omega t) + B \sin(\omega t)$	`\frac{d^2 y}{dt^2} + \omega^2 y = 0 \implies y(t) = A \cos(\omega t) + B \sin(\omega t)`

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Wave Equation (1D/3D) Hyperbolic wave propagation PDE	$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u = c^2 \frac{\partial^2 u}{\partial x^2}$	`\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u = c^2 \frac{\partial^2 u}{\partial x^2}`
Heat / Diffusion Equation Parabolic thermal diffusion PDE	$\frac{\partial u}{\partial t} = \alpha \nabla^2 u = \alpha \frac{\partial^2 u}{\partial x^2}$	`\frac{\partial u}{\partial t} = \alpha \nabla^2 u = \alpha \frac{\partial^2 u}{\partial x^2}`
Stokes' Theorem Surface curl to boundary circulation integral	$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_{S} (\nabla \times \vec{F}) \cdot d\vec{S}$	`\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_{S} (\nabla \times \vec{F}) \cdot d\vec{S}`
Divergence (Gauss) Theorem Volume divergence to surface flux integral	$\iint_{\partial V} \vec{F} \cdot d\vec{S} = \iiint_{V} (\nabla \cdot \vec{F}) \, dV$	`\iint_{\partial V} \vec{F} \cdot d\vec{S} = \iiint_{V} (\nabla \cdot \vec{F}) \, dV`

4. Linear Algebra, Matrix Operations & Vector Spaces

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
2×2 Parentheses Matrix 2x2 round matrix	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$	`\begin{pmatrix} a & b \\ c & d \end{pmatrix}`
3×3 Parentheses Matrix 3x3 round matrix	$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$	`\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}`
2×2 Square Bracket Matrix 2x2 square matrix	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	`\begin{bmatrix} a & b \\ c & d \end{bmatrix}`
3×3 Square Bracket Matrix 3x3 square matrix	$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$	`\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}`
2×2 Determinant 2x2 matrix determinant	$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a d - b c$	`\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a d - b c`
3×3 Determinant 3x3 matrix determinant	$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$	`\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}`
Column Vector (3D) 3D column vector	$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$	`\begin{pmatrix} x \\ y \\ z \end{pmatrix}`
Row Vector (3D) 1x3 row vector	$\begin{pmatrix} x & y & z \end{pmatrix}$	`\begin{pmatrix} x & y & z \end{pmatrix}`
Dot Product Scalar dot product	$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\theta = u_x v_x + u_y v_y + u_z v_z$	`\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos\theta = u_x v_x + u_y v_y + u_z v_z`

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Cross Product Vector 3D cross product	$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$	`\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}`
2×2 Matrix Inverse 2x2 matrix inversion	$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$	`A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}`
Eigenvalue & Eigenvector Equation Characteristic matrix equation	$A \vec{v} = \lambda \vec{v} \iff \det(A - \lambda I) = 0$	`A \vec{v} = \lambda \vec{v} \iff \det(A - \lambda I) = 0`
Singular Value Decomposition (SVD) Matrix factorization into singular components	$A = U \Sigma V^{T}$	`A = U \Sigma V^{T}`

5. Trigonometry, Analytic Identities & Logarithmic Functions

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Pythagorean Trig Identity Core trigonometric identity	$\sin^2(\theta) + \cos^2(\theta) = 1$	`\sin^2(\theta) + \cos^2(\theta) = 1`
Tangent-Secant Identity Derived Pythagorean identities	$1 + \tan^2(\theta) = \sec^2(\theta), \quad 1 + \cot^2(\theta) = \csc^2(\theta)$	`1 + \tan^2(\theta) = \sec^2(\theta), \quad 1 + \cot^2(\theta) = \csc^2(\theta)`
Sine Angle Sum & Difference Sine addition theorem	$\sin(\alpha \pm \beta) = \sin\alpha \cos\beta \pm \cos\alpha\sin\beta$	`\sin(\alpha \pm \beta) = \sin\alpha \cos\beta \pm \cos\alpha\sin\beta`
Cosine Angle Sum & Difference Cosine addition theorem	$\cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha\sin\beta$	`\cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha\sin\beta`
Double Angle Sine Double angle expansion	$\sin(2\theta) = 2 \sin\theta \cos\theta$	`\sin(2\theta) = 2 \sin\theta \cos\theta`
Double Angle Cosine Double angle cosine expansions	$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$	`\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta`
Law of Sines Triangular ratio of sides to opposite sines	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2 R$	`\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2 R`
Law of Cosines Generalization of Pythagorean theorem	$c^2 = a^2 + b^2 - 2 a b \cos C$	`c^2 = a^2 + b^2 - 2 a b \cos C`

6. Classical, Electromagnetic, Thermodynamic & Quantum Physics

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<thead>
<tr><th>CONCEPT / FORMULA NAME</th><th>RENDERED MATHEMATICAL OUTPUT</th><th>LATEX CODE</th></tr>
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<tbody>
<tr><td>Newton's Second Law
Force equals mass times acceleration</td><td>$\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$</td><td><code>\vec{F} = m \vec{a} = \frac{d\vec{p}}{dt}</code></td></tr>
<tr><td>Universal Gravitation
Newton's universal law of gravitation</td><td>$F = G\frac{m_1 m_2}{r^2}$</td><td><code>F = G \frac{m_1 m_2}{r^2}</code></td></tr>
<tr><td>Kinematic Motion Equations
Constant acceleration equations of motion</td><td>$v = v_0 + at, \quad x = x_0 + v_0 t + \frac{1}{2}at^2, \quad v^2 = v_0^2 + 2a(x - x_0)$</td><td><code>v = v_0 + a t, \quad x = x_0 + v_0 t + \frac{1}{2}a t^2, \quad v^2 = v_0^2 + 2a(x - x_0)</code></td></tr>
<tr><td>Kinetic Energy & Work
Work-energy theorem</td><td>$E_k = \frac{1}{2}mv^2, \quad W = \int \vec{F} \cdot d\vec{r} = \Delta E_k$</td><td><code>E_k = \frac{1}{2} m v^2, \quad W = \int \vec{F} \cdot d\vec{r} = \Delta E_k</code></td></tr>
<tr><td>Gravitational Potential Energy
Potential energy near surface of Earth</td><td>$U = mgh$</td><td><code>U = m g h</code></td></tr>
<tr><td>Hooke's Law & Elastic Potential
Spring restorative force and period</td><td>$F = -kx, \quad U_s = \frac{1}{2}kx^2, \quad T = 2\pi\sqrt{\frac{m}{k}}$</td><td><code>F = -k x, \quad U_s = \frac{1}{2} k x^2, \quad T = 2\pi \sqrt{\frac{m}{k}}</code></td></tr>
<tr><td>Mass-Energy Equivalence
Einstein's theory of special relativity</td><td>$E = mc^2 = \sqrt{(pc)^2 + (m_0 c^2)^2}$</td><td><code>E = m c^2 = \sqrt{(p c)^2 + (m_0 c^2)^2}</code></td></tr>
<tr><td>Coulomb's Law
Electrostatic force between point charges</td><td>$F = k_e \frac{|q_1 q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1 q_2|}{r^2}$</td><td><code>F = k_e \frac{|q_1 q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1 q_2|}{r^2}</code></td></tr>
<tr><td>Ohm's Law & Electric Power
Voltage, current, resistance, and electrical power</td><td>$V = IR, \quad P = IV = I^2 R = \frac{V^2}{R}$</td><td><code>V = I R, \quad P = I V = I^2 R = \frac{V^2}{R}</code></td></tr>
<tr><td>Lorentz Force Law
Force on charged particle in electromagnetic fields</td><td>$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$</td><td><code>\vec{F} = q \left(\vec{E} + \vec{v} \times \vec{B} \right)</code></td></tr>
</tbody>
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Faraday's Induction Law Electromotive force induced by changing magnetic flux	$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\iint \vec{B} \cdot d\vec{A}$	`\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\iint \vec{B} \cdot d\vec{A}`
Bernoulli's Fluid Equation Conservation of energy in fluid dynamics	$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$	`P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}`
Schrödinger Wave Equation Fundamental quantum state time evolution equation	$i\hbar \frac{\partial}{\partial t}\Psi(\vec{r}, t) = \hat{H}\Psi(\vec{r}, t)$	`i\hbar \frac{\partial}{\partial t}\Psi(\vec{r}, t) = \hat{H}\Psi(\vec{r}, t)`
Heisenberg Uncertainty Principle Quantum position and momentum uncertainty	$\Delta x \, \Delta p \ge \frac{\hbar}{2}$	`\Delta x \, \Delta p \ge \frac{\hbar}{2}`
Planck-Einstein Relation Photon energy as function of frequency	$E = h \nu = \hbar \omega = \frac{h c}{\lambda}$	`E = h \nu = \hbar \omega = \frac{h c}{\lambda}`
de Broglie Wavelength Wave-particle duality matter wavelength	$\lambda = \frac{h}{p} = \frac{h}{m v}$	`\lambda = \frac{h}{p} = \frac{h}{m v}`
Relativistic Time Dilation Time dilation at relativistic velocity	$\Delta t' = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma \Delta t$	`\Delta t' = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma \Delta t`
Snell's Law of Refraction Optics light refraction at refractive interface	$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$	`n_1 \sin(\theta_1) = n_2 \sin(\theta_2)`
Thin Lens / Lensmaker Equation Focal length and optical magnification	$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}, \quad M = -\frac{d_i}{d_o}$	`\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}, \quad M = -\frac{d_i}{d_o}`
First Law of Thermodynamics Conservation of internal energy, heat, and work	$\Delta U = Q - W$	`\Delta U = Q - W`
Boltzmann's Entropy Formula Statistical definition of entropy	$S = k_B \ln(\Omega)$	`S = k_B \ln(\Omega)`

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Maxwell's Equations (Full Set) Complete classical electromagnetism system	$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} = -\frac{\partial\vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t}$	`\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} = -\frac{\partial\vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t}`

7. Stoichiometry, Chemical Kinetics, Thermodynamics & Electrochemistry

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Ideal Gas Law Equation of state of ideal gas	$P V = n R T = N k_B T$	`P V = n R T = N k_B T`
Van der Waals Real Gas Equation of state of real gas	$\left(P + \frac{a n^2}{V^2}\right) (V - n b) = n R T$	`\left(P + \frac{a n^2}{V^2}\right) (V - n b) = n R T`
Gibbs Free Energy Spontaneity criterion for chemical reactions	$\Delta G = \Delta H - T \Delta S = -R T \ln(K)$	`\Delta G = \Delta H - T \Delta S = -R T \ln(K)`
Arrhenius Rate Equation Reaction rate constant temperature dependence	$k = A e^{-\frac{E_a}{R T}}$	`k = A e^{-\frac{E_a}{R T}}`
Nernst Equation (Electrochemistry) Cell potential under non-standard conditions	$E = E^\circ - \frac{R T}{n F} \ln(Q) = E^\circ - \frac{0.0592}{n} \log_{10}(Q)$	`E = E^\circ - \frac{R T}{n F} \ln(Q) = E^\circ - \frac{0.0592}{n} \log_{10}(Q)`
Henderson-Hasselbalch Equation Buffer solution pH calculation	$\text{pH} = \text{p}K_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]} \right)$	`\text{pH} = \text{p}K_a + \log_{10}\left( \frac{[\text{A}^-]}{[\text{HA}]} \right)`
Beer-Lambert Law Spectrophotometric light absorption law	$A = \varepsilon \cdot c \cdot l = -\log_{10}\left(\frac{I}{I_0} \right)$	`A = \varepsilon \cdot c \cdot l = -\log_{10}\left( \frac{I}{I_0} \right)`
Equilibrium Constant (Keq) Law of mass action equilibrium ratio	$K_{eq} = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$	`K_{eq} = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Chemical Rate Law Reaction rate order dependence	$r = k [\text{A}]^m [\text{B}]^n$	`r = k [\text{A}]^m [\text{B}]^n`
Photosynthesis Reaction Biological light-driven carbon fixation	$6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	`6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2`
Cellular Respiration Reaction Aerobic metabolism ATP generation	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 36\text{ATP}$	`\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 36 \text{ATP}`

8. Population Genetics, Enzymology, Ecology, Physiology & Biophysics

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Hardy-Weinberg Equilibrium Population genetics allele frequency distribution	$p^2 + 2pq + q^2 = 1 \quad \text{and} \quad p + q = 1$	`p^2 + 2pq + q^2 = 1 \quad \text{and} \quad p + q = 1`
Michaelis-Menten Kinetics Enzyme catalysis velocity reaction model	$v = \frac{V_{\max} [S]}{K_m + [S]}$	`v = \frac{V_{\max} [S]}{K_m + [S]}`
Lineweaver-Burk Double Reciprocal Linearized enzyme kinetics transformation	$\frac{1}{v} = \frac{K_m}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}$	`\frac{1}{v} = \frac{K_m}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}`
Exponential Population Growth Malthusian unconstrained population growth	$\frac{dN}{dt} = r N \implies N(t) = N_0 e^{r t}$	`\frac{dN}{dt} = r N \implies N(t) = N_0 e^{r t}`
Logistic Population Growth Carrying capacity K constrained growth model	$\frac{dN}{dt} = r N \left(1 - \frac{N}{K} \right)$	`\frac{dN}{dt} = r N \left( 1 - \frac{N}{K} \right)`
Shannon-Wiener Diversity Index Ecological community biodiversity measurement	$H' = -\sum_{i=1}^{S} p_i \ln(p_i)$	`H' = -\sum_{i=1}^{S} p_i \ln(p_i)`
Goldman-Hodgkin-Katz (GHK) Voltage Cell membrane resting potential across ion permeabilities	$V_m = \frac{R T}{F} \ln\left(\frac{P_{\text{K}}[\text{K}^+]_o + P_{\text{Na}}[\text{Na}^+]_o + P_{\text{Cl}}[\text{Cl}^-]_i}{P_{\text{K}}[\text{K}^+]_i + P_{\text{Na}}[\text{Na}^+]_i + P_{\text{Cl}}[\text{Cl}^-]_o} \right)$	`V_m = \frac{R T}{F} \ln\left( \frac{P_{\text{K}}[\text{K}^+]_o + P_{\text{Na}}[\text{Na}^+]_o + P_{\text{Cl}}[\text{Cl}^-]_i}{P_{\text{K}}[\text{K}^+]_i + P_{\text{Na}}[\text{Na}^+]_i + P_{\text{Cl}}[\text{Cl}^-]_o} \right)`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Lotka-Volterra Predator-Prey Nonlinear biological predator-prey dynamics	$\frac{dx}{dt} = \alpha x - \beta x y, \quad \frac{dy}{dt} = \delta x y - \gamma y$	`\frac{dx}{dt} = \alpha x - \beta x y, \quad \frac{dy}{dt} = \delta x y - \gamma y`
Kleiber's Law (Metabolic Scaling) Allometric scaling law of basal metabolic rate to mass	$B = B_0 \, M^{\frac{3}{4}}$	`B = B_0 \, M^{\frac{3}{4}}`

9. Probability, Inferential Statistics & Artificial Intelligence / Machine Learning

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Bayes' Theorem Fundamental posterior probability inversion	$P(A \mid B) = \frac{P(B \mid A) \, P(A)}{P(B)} = \frac{P(B \mid A) \, P(A)}{\sum_{i} P(B \mid A_i) P(A_i)}$	`P(A|B) = \frac{P(B|A) \, P(A)}{P(B)} = \frac{P(B|A) \, P(A)}{\sum_{i} P(B|A_i) P(A_i)}`
Gaussian Normal Distribution Continuous probability density function	$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	`f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}`
Sample Variance & Std Deviation Unbiased estimator of variance	$s^2 = \frac{1}{n - 1} \sum_{i=1}^{n} (x_i - \bar{x})^2, \quad s = \sqrt{s^2}$	`s^2 = \frac{1}{n - 1} \sum_{i=1}^{n} (x_i - \bar{x})^2, \quad s = \sqrt{s^2}`
Pearson Correlation (r) Linear bivariate correlation coefficient	$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$	`r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}`
Binomial Probability Mass (PMF) Probability of k successes in n trials	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$	`P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}`
Poisson Distribution PMF Probability of count occurrences per interval	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$	`P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}`
Sigmoid / Logistic Activation Machine learning logistic activation function	$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{e^z + 1}$	`\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{e^z + 1}`
Softmax Probability Function Multi-class probability distribution normalization	$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$	`\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}`
Cross-Entropy Loss Deep learning classification loss function	$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,k} \log(\hat{y}_{i,k})$	`\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,k} \log(\hat{y}_{i,k})`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Mean Squared Error (MSE) Regression mean squared deviation loss	$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$	`\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2`
Chi-Square Test Statistic Goodness of fit statistical test	$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$	`\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}`

10. Mathematical Operators, Relations, Sets, Logic & Arrows

Symbol	Command	Description
$\pm$	`\pm`	Plus-minus
$\mp$	`\mp`	Minus-plus
$\times$	`\times`	Multiplication cross
$\div$	`\div`	Division sign
$\cdot$	`\cdot`	Center dot product
$\circ$	`\circ`	Function composition
$\ast$	`\ast`	Asterisk / convolution
$\oplus$	`\oplus`	Direct sum
$\otimes$	`\otimes`	Tensor product
$\odot$	`\odot`	Hadamard product
$\dagger$	`\dagger`	Hermitian conjugate
$=$	`=`	Equals
$\neq$	`\neq`	Not equal
$\approx$	`\approx`	Approximately equal
$\equiv$	`\equiv`	Identical / congruent
$\sim$	`\sim`	Similar / distributed as
$\propto$	`\propto`	Proportional to
$<$	`<`	Less than
$>$	`>`	Greater than
$\leq$	`\leq`	Less than or equal
$\geq$	`\geq`	Greater than or equal
$\ll$	`\ll`	Much less than
$\gg$	`\gg`	Much greater than
$\cong$	`\cong`	Isomorphic / congruent
$\in$	`\in`	Element of
$\notin$	`\notin`	Not element of
$\subset$	`\subset`	Proper subset
$\subseteq$	`\subseteq`	Subset or equal
$\not\subset$	`\not\subset`	Not a subset
$\supset$	`\supset`	Superset
$\supseteq$	`\supseteq`	Superset or equal
$\cup$	`\cup`	Set union
$\cap$	`\cap`	Set intersection
$\emptyset$	`\emptyset`	Empty set
$\setminus$	`\setminus`	Set difference
$\forall$	`\forall`	For all
$\exists$	`\exists`	There exists
$\nexists$	`\nexists`	There does not exist
$\neg$	`\neg`	Logical NOT
$\land$	`\land`	Logical AND
$\lor$	`\lor`	Logical OR
$\therefore$	`\therefore`	Therefore
$\because$	`\because`	Because / since
$\mathbb{R}$	`\mathbb{R}`	Real numbers
$\mathbb{C}$	`\mathbb{C}`	Complex numbers
$\mathbb{N}$	`\mathbb{N}`	Natural numbers
$\mathbb{Z}$	`\mathbb{Z}`	Integers
$\mathbb{Q}$	`\mathbb{Q}`	Rational numbers
$\rightarrow$	`\rightarrow`	Right arrow
$\leftarrow$	`\leftarrow`	Left arrow
$\leftrightarrow$	`\leftrightarrow`	Bidirectional arrow
$\Rightarrow$	`\Rightarrow`	Implies
$\Leftarrow$	`\Leftarrow`	Is implied by
$\Leftrightarrow$	`\Leftrightarrow`	If and only if (IFF)
$\mapsto$	`\mapsto`	Maps to
$\uparrow$	`\uparrow`	Up arrow
$\downarrow$	`\downarrow`	Down arrow
∞	`\infty`	Infinity
∂	`\partial`	Partial differential
∇	`\nabla`	Nabla / Del gradient
$\hbar$	`\hbar`	Reduced Planck constant
ℓ	`\ell`	Script l
$\aleph$	`\aleph`	Aleph cardinal
$\angle$	`\angle`	Angle
$\perp$	`\perp`	Perpendicular
$\parallel$	`\parallel`	Parallel
$\dots$	`\dots`	Horizontal ellipsis
$\vdots$	`\vdots`	Vertical ellipsis
$\ddots$	`\ddots`	Diagonal ellipsis

11. The Complete Greek Alphabet (Lowercase & Uppercase)

α `\alpha` alpha (Lower)	β `\beta` beta (Lower)	γ `\gamma` gamma (Lower)	δ `\delta` delta (Lower)	ε `\epsilon` epsilon (Lower)	ε `\varepsilon` varepsilon (Lower)
ζ `\zeta` zeta (Lower)	η `\eta` eta (Lower)	θ `\theta` theta (Lower)	ϑ `\vartheta` vartheta (Lower)	ι `\iota` iota (Lower)	κ `\kappa` kappa (Lower)
λ `\lambda` lambda (Lower)	μ `\mu` mu (Lower)	ν `\nu` nu (Lower)	ξ `\xi` xi (Lower)	π `\pi` pi (Lower)	ϖ `\varpi` varpi (Lower)
ρ `\rho` rho (Lower)	ϱ `\varrho` varrho (Lower)	σ `\sigma` sigma (Lower)	ς `\varsigma` varsigma (Lower)	τ `\tau` tau (Lower)	υ `\upsilon` upsilon (Lower)
φ `\phi` phi (Lower)	ϕ `\varphi` varphi (Lower)	χ `\chi` chi (Lower)	ψ `\psi` psi (Lower)	ω `\omega` omega (Lower)	Γ `\Gamma` Gamma (Upper)
Δ `\Delta` Delta (Upper)	Θ `\Theta` Theta (Upper)	Λ `\Lambda` Lambda (Upper)	Ξ `\Xi` Xi (Upper)	Π `\Pi` Pi (Upper)	Σ `\Sigma` Sigma (Upper)
Υ `\Upsilon` Upsilon (Upper)	Φ `\Phi` Phi (Upper)	Ψ `\Psi` Psi (Upper)	Ω `\Omega` Omega (Upper)		